



Unit 5 - BCT: Analyzing Blockchain Tech Challenges and Solutions

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TECHNOLOGICAL ISSUES:

A number of technical challenges related to the blockchain, whether a specific one or the model in general, have been identified. Some think that the defacto standard will be the Bitcoin blockchain, as it is the incumbent, with the most widely deployed infrastructure and such network effects that it cannot help but be the standardized base. Others are building different new and separate blockchains (like Ethereum) or technology that does not use a blockchain (like Ripple).

One central challenge with the underlying Bitcoin technology is scaling up from the current maximum limit of 7 transactions per second (the VISA credit card processing network routinely handles 2,000 transactions per second and can accommodate peak volumes of 10,000 transactions per second), especially if there were to be mainstream adoption of Bitcoin.

Some of the other issues that are needed to be considered while adopting the Blockchain Technologies are:

➤ **THROUGHPUT:**

- The Bitcoin network has a potential issue with throughput in that it is processing only one transaction per second (tps), with a theoretical current maximum of 7 tps.
- Core developers maintain that this limit can be raised when it becomes necessary.
- One way that Bitcoin could handle higher throughput is if each block were bigger, though right now that leads to other issues with regard to size and blockchain bloat.
- Comparison metrics in other transaction processing networks are VISA (2,000 tps typical; 10,000 tps peak), Twitter (5,000 tps typical; 15,000 tps peak), and advertising networks (>100,000 tps typical).

➤ **LATENCY:**

- Right now, each Bitcoin transaction block takes 10 minutes to process, meaning that it can take at least 10 minutes for your transaction to be confirmed.
- For sufficient security, you should wait more time—about an hour—and for larger transfer amounts it needs to be even longer, because it must outweigh the cost of a double-spend attack (in which Bitcoins are double-spent in a separate transaction before the merchant can confirm their reception in what appears to be the intended transaction).
- Again, as the comparison metric, VISA takes seconds at most.

➤ **SIZE AND BANDWIDTH:**

- The blockchain is 25 GB, and grew by 14 GB in the last year. So, it already takes a long time to download (e.g., 1 day). If throughput were to increase by a factor of 2,000 to VISA standards, for example, that would be 1.42 PB/year or 3.9 GB/day. At 150,000 tps, the blockchain would grow by 214 PB/year.
- The Bitcoin community calls the size problem “bloat,” but that assumes that we want a small blockchain; however, to really scale to mainstream use, the blockchain would need to be big, just more efficiently accessed.
- This motivates centralization, because it takes resources to run the full node, and only about 7,000 servers worldwide do in fact run full Bitcoin nodes, meaning the Bitcoin daemon (the full Bitcoin node running in the background). It is being discussed whether locations running full nodes should be compensated with rewards.
- Although 25 GB of data is trivial in many areas of the modern “big data” era and data-intensive science with terabytes of data being the standard, this data can be compressed, whereas the blockchain cannot for security and accessibility reasons. However, perhaps this is an opportunity to innovate new kinds of compression algorithms that would make the blockchain (at much larger future scales) still usable, and storable, while retaining its integrity and accessibility.

- One innovation to address blockchain bloat and make the data more accessible is APIs, like those from Chain and other vendors, that facilitate automated calls to the full Bitcoin blockchain.
- Some of the operations are to obtain address balances and balances changes, and notify user applications when new transactions or blocks are created on the network.

➤ **SECURITY:**

- There are some potential security issues with the Bitcoin blockchain. The most worrisome is the possibility of a 51-percent attack, in which one mining entity could grab control of the blockchain and double-spend previously transacted coins into his own account.
- The issue is the centralization tendency in mining where the competition to record new transaction blocks in the blockchain has meant that only a few large mining pools control the majority of the transaction recording.
- At present, the incentive is for them to be good players, and some (like Ghash.io) have stated that they would not take over the network in a 51-percent attack, but the network is insecure.
- Double-spending might also still be possible in other ways—for example, spoofing users to resend transactions, allowing malicious coders to double-spend coins.
- Another security issue is that the current cryptography standard that Bitcoin uses, Elliptic Curve Cryptography, might be crackable as early as 2015; however, financial cryptography experts have proposed potential upgrades to address this weakness.

➤ **WASTED RESOURCES:**

- Mining draws an enormous amount of energy, all of it wasted.
- The earlier estimate cited was \$15 million per day, and other estimates are higher.
- On one hand, it is the very wastefulness of mining that makes it trustable—that rational agents compete in an otherwise useless proof-of-work effort in hopes of the possibility of reward—but on the other hand, these spent resources have no benefit other than mining.

➤ **USABILITY:**

- The API for working with Bitcoin (the full node of all code) is far less user-friendly than the current standards of other easy-to-use modern APIs, such as widely used REST APIs.

➤ **VERSIONING, HARD FORKS, MULTIPLE CHAINS:**

- One issue is the proliferation of blockchains, and that with so many different blockchains in existence, it could be easy to deploy the resources to launch a 51-percent attack on smaller chains.
- Another issue is that when chains are split for administrative or versioning purposes, there is no easy way to merge or cross-transact on forked chains.

PREFERABLE SOLUTIONS:

Another significant technical challenge and requirement is that a full ecosystem of plug-and-play solutions be developed to provide the entire value chain of service delivery. For example, linked to the blockchain there needs to be secure decentralized storage (MaidSafe, Storj), messaging, transport, communications protocols, name-space and address management, network administration, and archival. Ideally, the blockchain industry would develop similarly to the cloud-computing model, for which standard infrastructure components—like cloud servers and transport systems—were defined and implemented very quickly at the beginning to allow the industry to focus on the higher level of developing value-added services instead of the core infrastructure. This is particularly important in the blockchain economy due to the sensitive and complicated cryptographic engineering aspects of decentralized networks.

The industry is sorting out exactly how much computer network security, cryptography, and mathematics

expertise the average blockchain startup should have—ideally not much if they can rely on a secure infrastructure stack on which this functionality already exists.

That way, the blockchain industry's development can be hastened, without every new business having to reinvent the wheel and worry about the fact that its first customer-facing wallet was not multisig (or whatever the current industry standard is, as cryptographic security standards will likely continue to iterate).

Some of the partial proposed solutions to the technical issues discussed here are as follows:

➤ **OFFLINE WALLETS TO STORE THE MAJORITY OF COINS:**

Different manner of offline wallets could be used to store the bulk of consumer crypto coins. **For example**, paper wallets, cold storage, and bit cards.

➤ **DARK POOLS:**

There could be a more granular value chain such that big crypto-exchanges operate their own internal databases of transactions, and then periodically synchronize a summary of the transactions with the blockchain—an idea borrowed from the banking industry.

➤ **ALTERNATIVE HASHING ALGORITHMS:**

Litecoin and other cryptocurrencies use scrypt, which is at least slightly faster than Bitcoin, and other hashing algorithms could be innovated.

➤ **ALTERNATIVES TO PROOF OF WORK FOR BYZANTINE CONSENSUS:**

There are many other consensus models proposed—such as proof of stake, hybrids, and variants—that have lower latency, require less computational power, waste fewer resources, and improve security for smaller chains. Consensus without mining is another area being explored, such as in Tendermint's modified version of DLS (the solution to the Byzantine Generals' Problem by Dwork, Lynch, and Stockmeyer), with bonded coins belonging to byzantine participants. Another idea for consensus without mining or proof of work is through a consensus algorithm such as Hyperledger's, which is based on the Practical Byzantine Fault Tolerance algorithm.

➤ **ONLY FOCUS ON THE MOST RECENT OR UNSPENT OUTPUTS:**

Many blockchain operations could be based on surface calculations of the most recent or unspent outputs, similar to how credit card transactions operate. "Thin wallets" operate this way, as opposed to querying a full Bitcoin node, and this is how Bitcoin wallets work on cellular telephones. A related proposal is Cryptonite, which has a "mini-blockchain" abbreviated data scheme.

➤ **BLOCK CHAIN INTEROPERABILITY:**

To coordinate transactions between blockchains, there are several side chains projects proposed, such as those by Blockstream.

➤ **POSTING BOND DEPOSITORS:**

The security of proposed alternative consensus mechanisms like Tendermint's DLS protocol (which requires no proof-of-work mining) could be reinforced with structural elements such as requiring miners to post bond deposits to blockchains. This could help resolve the security issue of the "nothing at stake in short time ranges" problem, where malicious players (before having a stake) could potentially fork the blockchain and steal cryptocurrency in a double-spend attack. Bond deposits could be posted to blockchains like Tendermint does, making it costly to fork and possibly improving operability and security.

➤ **REST APIS:**

Essentially secure calls in real time, these could be used in specific cases to help usability. Many blockchain companies provide alternative wallet interfaces that have this kind of functionality, such as Blockchain.info's numerous wallet APIs.

BUSINESS MODEL CHALLENGES:

- Another noted challenge, both functional and technical, is related to business models.
- At first traditional business models might not seem applicable to Bitcoin since the whole point of decentralized peer-to-peer models is that there are no facilitating intermediaries to take a cut/transaction fee (as in one classical business model).
- However, there are still many worthwhile revenue-generating products and services to provide in the new blockchain economy.
- Education and mainstream user-friendly tools are obvious low-hanging fruit (for example, being targeted by Coinbase, Circle Internet Financial, and Xapo), as is improving the efficiency of the entire worldwide existing banking and finance infrastructure like Ripple another almost “no brainer” project, when blockchain principles are understood.
- Looking ahead, reconfiguring all of business and commerce with smart contracts in the Bitcoin 2.0 era could likely be complicated and difficult to implement, with many opportunities for service providers to offer implementation services, customer education, standard setting, and other value-added facilitations.
- Some of the many types of business models that have developed with enterprise software and cloud computing might be applicable, too, for the Bitcoin economy—for example, the Red Hat model (fee-based services to implement opensource software), and SaaS, providing Software as a Service, including with customization.
- One possible job of the future could be smart contract auditor, to confirm that AI smart contracts running on the blockchain are indeed doing as instructed, and determining and measuring how the smart contracts have self-rewritten to maximize the issuing agent’s utility.

SCADALS AND PUBLIC PERCEPTION:

- One of the biggest barriers to further Bitcoin adoption is its public perception as a venue for (and possible abettor of) the dark net’s money-laundering, drug-related, and other illicit activity—for example, illegal goods online marketplaces such as SilkRoad.
- Bitcoin and the blockchain are themselves neutral, as any technology, and are “dual use”; that is, they can be used for good or evil. Although there are possibilities for malicious use of the blockchain, the potential benefits greatly outweigh the potential downsides.
- Over time, public perception can change as more individuals themselves have wallets and begin to use Bitcoin. Still, it must be acknowledged that Bitcoin as a pseudonymous enabler can be used to facilitate illegal and malicious activities, and this invites in-kind “Red Queen” responses (context-specific evolutionary arms races) appropriate to the blockchain.
- Computer virus detection software arose in response to computer viruses; and so far, some features of the same constitutive technologies of Bitcoin (like Tor, a free and open software network) have been deployed back into detecting malicious players.
- Another significant barrier to Bitcoin adoption is the ongoing theft, scandals, and scams (like so-called new altcoin “pump and dump” scams that try to bid up new alt-coins to quickly profit) in the industry.
- The collapse of the largest Bitcoin exchange at the time, Tokyo-based MtGox, in March 2014 came to wide public attention.

- An explanation is still needed for the confusing irony that somehow in the blockchain, the world's most public transparent ledger, coins can disappear and still remain lost months later.
- The company said it had been hacked, and that the fraud was a result of a problem known as a "transaction malleability bug." The bug allowed malicious users to double-spend, transferring Bitcoins into their accounts while making MtGox think the transfer had failed and thus repeat the transactions, in effect transferring the value twice.
- Analysts remain unsure if MtGox was an externally perpetrated hack or an internal embezzlement. The issue is that these kinds of thefts persist. **For example**, recent headlines inform us that the Moolah CEO disappeared with \$1.4 million in Bitcoin (October 2014), \$2 million of Vericoins was stolen (July 2014), and \$620,000 was stolen in a Dogecoin mining attack (June 2014).
- Blockchain industry models need to solidify and mature such that there are better safeguards in place to stabilize the industry and allow both insiders and outsiders to distinguish between good and bad players.
- Oversight need not come from outside; congruently decentralized vetting, confirmation, and monitoring systems within the ecosystem could be established.
- An analogy from citizen science is realizing that oversight functions are still important, and reinforce the system by providing checks and balances.
- In DIY genomics participant-organized research studies, for example, the oversight function is still fulfilled, but in some cases with a wholly new role relevant to the ecosystem independent citizen ethicists as opposed to traditional top-down overseers (in the form of a human-subjects research Institutional Review Board). Other self-regulating industries include movies, video games, and comic books.
- There is the possibility that the entire blockchain industry could just collapse (either due to already prognosticated problems or some other factor as yet unforeseen).
- The blockchain economy does have a strong presence, as measured by diverse metrics such as coin market capitalizations, investment in the sector, number of startups and people working in the sector, lines of GitHub code committed, and the amount of "newspaper ink" devoted to the sector.
- Already the blockchain industry is bigger and better established than the previous run at digital currencies (virtual-world currencies like the Second Life Linden dollar).
- However, despite the progress to date and lofty ideals of Bitcoin, maybe it is still too early for digital currency; maybe all of the right safeguards and structures are not yet in place for digital currencies to go fully mainstream (although Apple Pay, more than any other factor, may pave the way to full mainstream acceptance of digital currencies). Apple Pay could quite possibly be enough for the short term. It will be a long time before Bitcoin has the same user-friendly attributes of Apple Pay, such as latency of confirmation time.

GOVERNMENT REGULATION:

- How government regulation unfolds could be one of the most significant factors and risks in whether the blockchain industry will flourish into a mature financial services industry.
- In the United States, there could be federal- and state-level legislation; deliberations continue into a second comment period regarding a much-discussed New York Bit license.
- The New York Bit license could set the tone for worldwide regulation. On one hand, the Bitcoin

industry is concerned about the extremely broad, wide-reaching, and extraterritorial language of the license as currently proposed.

- The license would encompass anyone doing anything with anyone else's Bitcoins, including basic wallet software (like the QT wallet).
- However, on the other hand, regulated consumer protections for Bitcoin industry participants, like KYC (know your customer) requirements for money service businesses (MSBs), could hasten the mainstream development of the industry and eradicate consumer worry of the hacking raids that seem to plague the industry.
- The deliberations and early rulings of worldwide governments on Bitcoin raise some interesting questions.

One issue is the potential practical impossibility of carrying out taxation with current methods.

A decentralized peer-to-peer sharing economy of Airbnb 2.0 and Uber 2.0 run on local implementations of OpenBazaar with individuals paying with cryptocurrencies renders traditional taxation structures impossible. The usual tracking and chokehold points to trace the consumption of goods and services might be gone. This has implications both for taxation and for the overall measurement of economic performance such as GDP calculations, which could have the beneficial impact of drawing populaces away from being overly and possibly incorrectly focused on consumption as a wellness metric. Instead, there could be an overhaul of the taxation system to a consumption-based tax on large-ticket visible items such as hard assets (cars, houses).

Chokehold points would need to be easily visible for taxation, a "tax on sight" concept. A potential shift from an income tax-based system to a consumption tax-based system could be a significant change for societies.

A second issue that blockchain technology raises with regard to government regulation is the value proposition offered by governments and their business model.

Some argue that in the modern era of big data, governments are increasingly unable to keep up with their record-keeping duties of recording and archiving information and making data easily accessible. On this view, governments could become obsolete because they cannot fund themselves the traditional way—by raising taxes. Blockchain technology could potentially help solve both of these challenges, and could at minimum supplement and help governments do their own jobs better, eventually making classes of government-provided services redundant. Recording all of a society's records on the blockchain could obviate the need for entire classes of public service. This view starkly paints governments as becoming redundant with the democratization of government features of the blockchain.

However, just as there might be both centralized and decentralized models to coordinate our activities in the world, there could likely be roles for both traditional government and new forms of blockchain-based government.

- There might still be a role for traditional centralized governments, but they will need to become economically rationalized, with real value propositions that resonate with constituencies, shrink costs, and demonstrate effectiveness.
- There could be hybrid governments in the future, like other industries, where automation is the forcing function, and the best "worker" for the job is a human/algorithmic pairing.
- Perfunctory repetitive tasks are automated with blockchain registries and smart contracts, whereas government employees can move up the value chain.

PRIVACY CHALLENGES FOR PERSONAL RECORDS:

There are many issues to be resolved before individuals would feel comfortable storing their personal records in a decentralized manner with a pointer and possibly access via the blockchain. The potential privacy nightmare is that if all your data is online and the secret key is stolen or exposed, you have little recourse. In the current cryptocurrency architecture, there are many scenarios in which this might happen, just as today with personal and corporate passwords being routinely stolen or databases hacked—with broad but shallow consequences; tens of thousands of people deal with a usually minor inconvenience. If a thorough personal record is stolen, the implications could be staggering for an individual: identity theft to the degree that you no longer have your identity at all.